
VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION OF VISION–LANGUAGE MODELS

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ABSTRACT

Vision–Language Models (VLMs) are an increasingly general foundation for zero-shot goal-conditioned value functions across a broad range of tasks and domains, yet their frozen pre-trained representations still limit generalization and temporal reasoning. We introduce VITA, a general zero-shot value function learning method that strengthens both capabilities through test-time adaptation. At inference, a lightweight adaptation module is updated with a single gradient step on a meta-learned self-supervised loss, so that every test-time update improves value estimation. By updating sequentially over a trajectory, VITA encodes history directly into its parameters, addressing a temporal reasoning limitation that is common to frozen VLMs. To mitigate shortcut learning, we introduce a dissimilarity-based sampling strategy that selects semantically diverse segments of a trajectory during training. Trained in a single environment, VITA generalizes broadly to diverse out-of-distribution tasks, environments, and embodiments in real-world robotic manipulation, consistently outperforming the state-of-the-art zero-shot method built on autoregressive VLMs. We further show that VITA’s zero-shot value estimates transfer to reward shaping for offline reinforcement learning, yielding multi-task policies on the Meta-World benchmark that surpass those trained with the simulator’s fuzzy-logic dense rewards.

1 INTRODUCTION

Vision-Language Models (VLMs) have demonstrated strong generalization across a wide spectrum of tasks and domains by learning from large-scale, unstructured web data without human supervision (Radford et al., 2021; Alayrac et al., 2022). In contrast, despite significant advances in learning generalist policies for domains as varied as robotics (Brohan et al., 2023; Kim et al., 2024) and 3D virtual environments (Reed et al., 2022; SIMA Team et al., 2024), state-of-the-art methods have yet to achieve comparable breadth, largely because they rely on expert demonstrations (Ho & Ermon, 2016). World models (Ha & Schmidhuber, 2018; Hafner et al., 2021; Matsuo et al., 2022) have emerged as a general-purpose solution, enabling agents to learn in simulated environments across settings as diverse as robotics (Yang et al., 2023; Agarwal et al., 2025), autonomous driving (Hu et al., 2024), and video games (Bruce et al., 2024). Yet even as world models generate realistic visual trajectories in increasingly diverse, open-ended environments (Hughes et al., 2024; Silver & Sutton, 2025), a broad and still-open challenge remains: *how can agents effectively learn from videos at scale without relying on human supervision?*

One broad line of research explores policy learning directly from expert visual trajectories by inferring actions from visual transitions (Edwards et al., 2019; Baker et al., 2022; Schmidt & Jiang, 2024; Zhang et al., 2025a). These latent actions can be mapped to executable controls across a variety of deployment settings, although this remains challenging in practice. In parallel, a complementary body of work pursues a general-purpose *goal-conditioned value function*, using it as a zero-shot reward shaping and supervision signal for reinforcement learning (RL) and imitation learning, respectively (Chen et al., 2021; Ma et al., 2022). Within this framework, goal-conditioned value estimation can be posed broadly as predicting how far an agent has progressed toward completing an arbitrary task, given only visual observations and a natural language task description (Lee et al., 2021a; Ma et al., 2023b; 2024; Dashora et al., 2025).

Prior work has employed pre-trained contrastive VLMs (Radford et al., 2021) for zero-shot reward shaping across many tasks, leveraging the similarity between task descriptions and visual observations in a shared multimodal representation space to ground progress in semantic context (Ma et al., 2023b; Baumli et al., 2023; Rocamonde et al., 2024). However, these methods generally fail to capture the temporal context needed to disambiguate visually similar states that arise at different stages of a task (e.g., folding vs. unfolding a shirt). Autoregressive VLMs (Alayrac et al., 2022) instead incorporate temporal context by conditioning on the entire visual trajectory within the prompt, yet they inherit a bias toward monotonically increasing predictions from the chronologically ordered datasets used in pre-training (Ma et al., 2024). Across both families of VLM architectures, reliance on frozen pre-trained representations (Ma et al., 2023b; 2024; Baumli et al., 2023; Rocamonde et al., 2024; Du et al., 2023) for zero-shot prediction broadly limits generalization and temporal reasoning alike (Pătrăucean et al., 2023). To make value function learning more broadly generalizable, recent work has explored large-scale pre-training (Alakuijala et al., 2024; Ma et al., 2023b), domain-specific fine-tuning (Zhang et al., 2025b), and demo-based reward specification (Sontakke et al., 2023).

In this work, we introduce VITA, a general zero-shot value function learning method that strengthens both generalization and temporal reasoning through test-time adaptation, outperforming the state-of-the-art zero-shot approach (Ma et al., 2024) on real-world robotic manipulation tasks. Unlike prior methods that lean on large-scale pre-training or expert demonstrations to achieve breadth, VITA adapts online to both the semantic and temporal context of each trajectory, making it broadly applicable across tasks and settings. At inference, a lightweight adaptation module is updated in negligible time with a single gradient step on a meta-learned self-supervised loss (Sun et al., 2024), so that each test-time update improves value function estimation (Finn et al., 2017). By updating sequentially over a test trajectory, the value function estimator encodes trajectory history into its parameters (Sun et al., 2024), thereby addressing a temporal reasoning limitation shared broadly by prior work. To mitigate shortcut learning (Geirhos et al., 2020), we introduce a dissimilarity-based sampling strategy that encourages reliance on semantic cues, supported by empirical evidence. Our evaluation spans the core capabilities expected of a general value function estimator: generalization under distribution shifts, differentiation between expert and non-expert trajectories, and reward shaping for offline RL (Levine et al., 2020).

We summarize our key contributions as follows:

- We propose VITA, a general test-time adaptation method that enhances the generalization and temporal reasoning of contrastive VLMs for zero-shot value function estimation across diverse tasks, without requiring task-specific demonstrations or large-scale pre-training.
- Trained in a single environment, VITA generalizes broadly to diverse out-of-distribution tasks, environments, and embodiments in robotic manipulation, consistently outperforming the state-of-the-art zero-shot method (Ma et al., 2024).
- VITA’s zero-shot value estimates transfer to reward shaping, yielding offline RL policies on the Meta-World benchmark for multi-task learning (MT10) that surpass those trained with the simulation’s fuzzy-logic dense rewards (Yu et al., 2020).

2 PRELIMINARIES

2.1 GOAL-CONDITIONED VALUE FUNCTIONS

For any video trajectory, task progress estimation can be viewed as goal-conditioned value function estimation, in which the reward reflects task completion. We therefore cast learning a goal-conditioned value function, in general terms, as predicting the degree of task completion from vision–language representations. Formally, the value function is defined as $V : \mathcal{O} \times \mathcal{G} \rightarrow [0, 1]$, which maps an observation $o_t \in \mathcal{O}$ and a goal specification $g \in \mathcal{G}$ to a scalar indicating the predicted progress toward goal completion. We set $V(o_t; g) = 0$ at the start of a task and $V(o_t; g) = 1$ at its completion. Across a wide range of tasks, progress is commonly aligned with temporal position in expert demonstrations, under the assumption that such trajectories exhibit monotonically increasing progress toward goal completion (Lee et al., 2021a; Ma et al., 2024; Dashora et al., 2025). Given an expert trajectory $\tau = (o_1, \dots, o_T) \sim \pi_E$, temporal progress is defined using normalized timestep indices: $V^{\pi_E}(o_t; g) = \frac{t}{T}$ where T is the trajectory length. Temporal progress thus provides a general supervision signal for learning a goal-conditioned value function V .

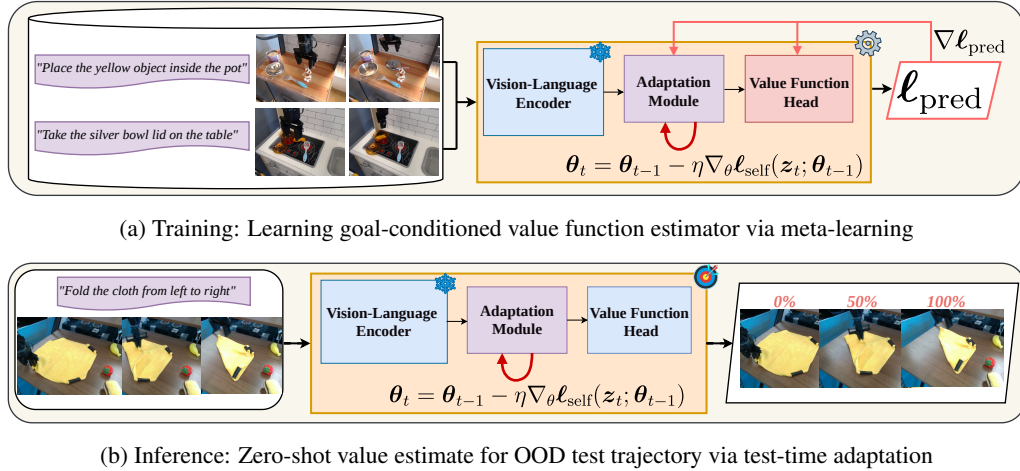


Figure 1: **Overview of VITA.** VITA learns a goal-conditioned value function through meta-learning and, via test-time adaptation, achieves zero-shot generalization to a broad range of out-of-distribution trajectories.

2.2 TEST-TIME TRAINING FOR SEQUENCE MODELING

Test-time training (TTT) (Sun et al., 2020) is a broadly applicable test-time adaptation method that treats inference as a self-supervised learning task (Goodfellow et al., 2016), updating model parameters on each test instance without access to labels. Although originally proposed for static image classification (Sun et al., 2020), the paradigm extends naturally to sequence modeling (Sun et al., 2024; Wang et al., 2023), where a parametric adaptation module f_{adapt} is updated at each timestep with a gradient step on a self-supervised loss. In contrast to standard sequence architectures that encode temporal history in activation states, such as hidden states (Hochreiter & Schmidhuber, 1997; Cho et al., 2014) or cached key-value pairs (Vaswani et al., 2017), TTT encodes history into the parameters of f_{adapt} through sequential updates at inference. The parameters of f_{adapt} thereby act as an implicit memory that encodes temporal history while preserving temporal order. Sun et al. (2024) further propose to meta-learn the self-supervised task, following the gradient-based meta-learning paradigm (Finn et al., 2017). In this setting (Sun et al., 2024), the self-supervised loss is parameterized by learnable linear projections and trained to minimize downstream prediction loss after a test-time update, rather than being minimized directly. In this work, we bring TTT to the more general problem of goal-conditioned value function learning: temporal context is captured through sequential updates, and the self-supervised task is meta-learned to improve value estimation rather than being predefined a priori.

3 VITA: ZERO-SHOT VALUE FUNCTIONS VIA TEST-TIME ADAPTATION

3.1 MODEL ARCHITECTURE

Our goal-conditioned value function estimator is a general, modular architecture built from three components: (1) a multimodal encoder that extracts joint visual-language representations from arbitrary visual trajectories and their task descriptions; (2) an adaptation module updated at test-time with a self-supervised loss that is meta-learned to improve value estimation; and (3) a regression head that predicts value estimates.

3.1.1 MULTIMODAL INPUT REPRESENTATION

We adopt a frozen contrastive vision-language encoder, CLIP (Radford et al., 2021), to extract representations from visual observations and goal descriptions; the design is agnostic to the specific encoder and applies to any such contrastive backbone. Given a visual trajectory $\tau = (o_1, \dots, o_T)$ and a language task description g , we concatenate their representations at each timestep t to form a sequence of joint multimodal representations $(z_1, z_2, \dots, z_T)$, where $z_t = [\phi_v(o_t); \phi_g(g)] \in \mathbb{R}^{2d}$,

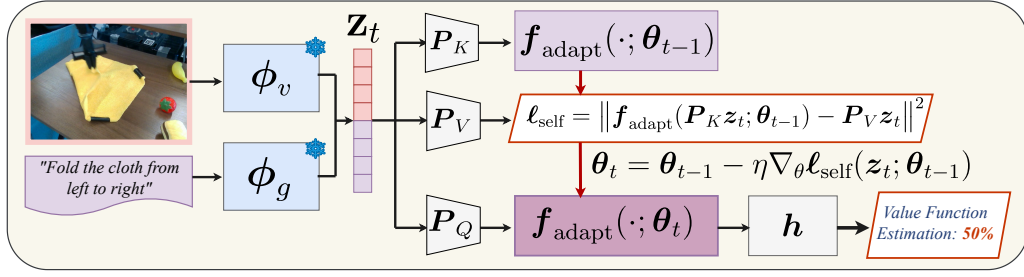


Figure 2: **Test-time Adaptation.** At inference, for each timestep t of an arbitrary trajectory, the adaptation module f_{adapt} is updated with a gradient step on a meta-learned self-supervised loss ℓ_{self} , progressively encoding temporal history.

and ϕ_v and ϕ_g denote the visual and language encoders, respectively. CLIP is pre-trained with a contrastive objective that aligns paired image-text inputs in a shared representation space (Radford et al., 2021). As a result, representations of visual observations that are semantically closer to goal completion tend to lie closer to the representation of the goal description.

3.1.2 TEST-TIME ADAPTATION

To adapt multimodal representations to both semantic and temporal context, we employ an adaptation module f_{adapt} that follows the general test-time training paradigm described in 2.2. At inference, the parameters of f_{adapt} are updated online via a gradient step on a meta-learned self-supervised loss ℓ_{self} . This loss takes the form of a reconstruction objective using learnable linear projections P_K and P_V , which are meta-learned during training to improve value estimation after test-time adaptation. In particular, given $P_K \in \mathbb{R}^{d' \times 2d}$ that generates a perturbed input view and $P_V \in \mathbb{R}^{d' \times 2d}$ for the target, the self-supervised loss is defined as:

$$\ell_{\text{self}}(z_t; \theta_{t-1}, P_K, P_V) = \|f_{\text{adapt}}(P_K z_t; \theta_{t-1}) - P_V z_t\|^2. \quad (1)$$

where θ_{t-1} denotes the parameters of f_{adapt} before test-time adaptation. At each timestep t , the parameters are updated by

$$\theta_t = \theta_{t-1} - \eta \nabla_{\theta} \ell_{\text{self}}(z_t; \theta_{t-1}), \quad (2)$$

with learning rate η . Through these sequential updates, f_{adapt} implicitly retains information from past visual observations, capturing temporal context in a manner that applies uniformly across trajectories of arbitrary length. In Section 4.6.2, we provide empirical evidence that incremental parameter updates of the value function estimator are broadly more effective than updates performed at the trajectory level.

3.1.3 ZERO-SHOT VALUE FUNCTION ESTIMATOR

After test-time adaptation, a meta-learned linear projection $P_Q \in \mathbb{R}^{d' \times 2d}$ maps the input z_t into an adaptation space $\mathbb{R}^{d'}$. Then, this representation is passed through the adaptation module f_{adapt} , followed by a regression head h that outputs predicted values of the goal-conditioned value function:

$$V(z_t; g) = h(f_{\text{adapt}}(P_Q z_t; \theta_t)) \quad (3)$$

The regression head h is a two-layer multilayer perceptron (MLP), following the value-function estimator architectures widely used in deep reinforcement learning (Espeholt et al., 2018). The network is trained with normalized progress labels y_t from expert demonstrations, as described in Section 2.1, using a supervised prediction loss ℓ_{pred} defined as the mean squared error between predicted values $V(z_t; g)$ and targets y_t . At inference, the regression head h stays frozen while the adaptation module f_{adapt} is updated online with the self-supervised objective. This lets the value function estimator generalize in a zero-shot setting to new tasks without requiring any task-specific demonstrations.

3.2 TRAINING PROCEDURE

We train the value function estimator with gradient-based meta-learning, optimizing a self-supervised loss ℓ_{self} so that test-time adaptation improves the supervised prediction loss ℓ_{pred} (Sun et al., 2024). Training operates over diverse sub-trajectories selected via dissimilarity-based sampling, a task-agnostic strategy that mitigates shortcut learning and encourages reliance on semantic and temporal cues.

3.2.1 META-LEARNING

During training, the adaptation module f_{adapt} is updated online at each timestep using the self-supervised loss ℓ_{self} . Following the general gradient-based meta-learning paradigm (Finn et al., 2017), we differentiate through these adaptation updates with respect to the supervised prediction loss ℓ_{pred} . This jointly optimizes the initialization θ_0 of f_{adapt} , the linear projections P_K, P_V, P_Q , and the regression head h . Formally, the total training loss combines the supervised prediction loss ℓ_{pred} with the self-supervised loss ℓ_{self} , weighted by a scalar λ . The learned parameters θ_0 then serve as the initialization for adaptation at inference time. As a result, the value function estimator learns a general ability to adapt its internal representations at test time so as to improve value function estimation on trajectories it has never seen.

3.2.2 DISSIMILARITY-BASED SAMPLING

Expert visual trajectories frequently contain consecutive frames that are highly redundant, a property observed broadly across video domains (Wang et al., 2018). Such redundancy can lead the value function estimator to exploit shortcut cues (Geirhos et al., 2020; Lee et al., 2021b) by overfitting to frequently occurring late-stage visual patterns. To counter this, we propose a general dissimilarity-based sampling strategy that constructs mini-batches from the most visually diverse sub-trajectories within each trajectory. This increases intra-batch variance and acts as a form of importance sampling that favors semantic diversity, independent of the particular task or domain.

Formally, given a sequence of multimodal representations $\{z_1, \dots, z_T\}$, we extract fixed-length sub-trajectories using a sliding window of size w_{tr} and stride s , yielding a candidate set $\mathcal{W}$. A diverse subset of size k maximizes pairwise dissimilarity:

$$\mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{\{w^i, w^j\} \in \binom{\mathcal{U}}{2}} \|w^i - w^j\|_2^2, \quad (4)$$

where each w^i is a sub-trajectory of length w_{tr} . Solving Eq. 4 is intractable, requiring enumeration of all $\binom{|\mathcal{W}|}{k}$ subsets. To approximate this objective efficiently, we adopt a scoring-based heuristic, assigning each window $w \in \mathcal{W}$ a score equal to its total dissimilarity with all others:

$$s(w) = \sum_{v \in \mathcal{W}} \|w - v\|_2^2, \quad \mathcal{W}' = \arg \max_{\mathcal{U} \subset \mathcal{W}, |\mathcal{U}|=k} \sum_{w \in \mathcal{U}} s(w). \quad (5)$$

Thus, by computing a diversity score s for each window and selecting the k highest-scoring windows, we obtain a heuristically diverse subset with polynomial-time complexity. The overhead is negligible relative to model training, as our complexity analysis in Appendix G shows, and the procedure scales gracefully to trajectories of varying length. We empirically validate dissimilarity-based sampling against full-trajectory sampling in the ablation study (Section 4.6.1), where it consistently improves the ability to distinguish expert from non-expert visual trajectories.

4 EXPERIMENTS

Our evaluation probes VITA along the core capabilities expected of a general goal-conditioned value function, spanning both real-world and simulated settings: (i) generalization in value estimation for real-world robotic manipulation under a wide range of distribution shifts in task, environment, and embodiment; (ii) differentiation between expert and non-expert visual trajectories in real-world robotic manipulation tasks; (iii) effective zero-shot reward shaping for offline RL in Meta-World MT10, a simulated benchmark for multi-task robot learning.

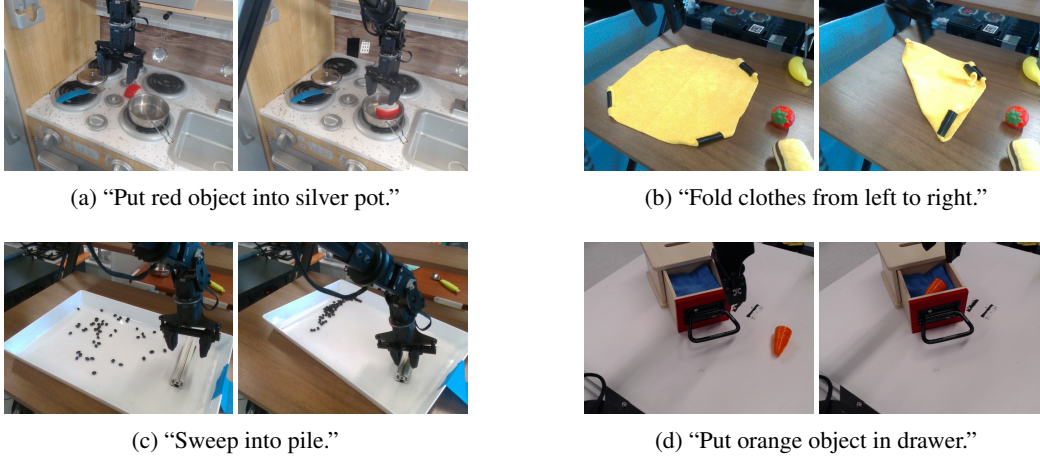


Figure 3: Representative visual trajectories paired with task descriptions, spanning the range of distribution shifts we consider. (a) In-distribution. (b, c) Environment shift. (d) Embodiment and environment shift.

4.1 TRAINING SETUP

We train VITA on expert visual trajectories paired with natural language task descriptions drawn from the BridgeData V2 dataset (Walke et al., 2023), which itself spans a wide range of manipulation tasks, environments, and robot embodiments. To stress-test generalization, we deliberately train on a narrow slice of this data: a curated subset of 2,986 expert demonstrations covering only pick-and-place manipulation, with no folding, sweeping, or stacking tasks. All demonstrations use a single robot embodiment (WidowX 250) across 4 configurations of the ToyKitchen environment. A further 287 expert demonstrations are held out as the in-distribution test set, referred to as `tk_pnp`. Appendix A includes examples from the training set. We use OpenCLIP ViT-B/32 as the frozen backbone for encoding video frames and task descriptions. The training objective combines ℓ_{pred} with ℓ_{self} , weighted by $\lambda_{\text{self}} = 0.5$. During training, we use dissimilarity-based sampling with window size $w_{\text{tr}} = 8$, number of sub-trajectories $k = 8$, and stride $s = 1$. At test time, we use one gradient step ($t_{\text{ep}} = 1$) with λ_{self} and learning rate $\eta = 0.1$. Further details and hyperparameter sweeps are reported in Appendix D.

4.2 BASELINES

We compare VITA against a broad set of value functions spanning both contrastive and autoregressive VLM families. VLM-CL computes zero-shot value estimates from cosine similarity between CLIP representations of each frame and task description (Baumli et al., 2023). VLM-RM regularizes CLIP embeddings by projecting features along the direction from a generic reference prompt to the task prompt (Rocamonde et al., 2024). CLIP-FT trains a supervised regression head on frozen CLIP multimodal representations. CLIP-GRU trains a Gated Recurrent Unit (GRU) (Cho et al., 2014) on frozen CLIP multimodal representations. GVL (Ma et al., 2024) is the state-of-the-art zero-shot value function that leverages autoregressive VLMs. In our experiments, GVL-0S refers to the zero-shot setting, where the VLM is prompted with only the test trajectory and task description. GVL-1S corresponds to a one-shot in-context setting, where a full shuffled trajectory from the training distribution, along with its progress labels, is provided as an example. We follow GVL in using Gemini 1.5 Pro (`gemini-1.5 Pro-latest`) (Gemini Team et al., 2024) with its proposed prompt template. To probe the generality of this baseline across model choices, we also tested Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias, and GPT-4o (Hurst et al., 2024), which declined to produce value estimates. Additional implementation details are provided in Appendix D.

Table 1: VOC scores for value function estimation across a broad range of distribution shifts. **ID** = In-Distribution, **ES** = Environment Shift, **EM** = Embodiment Shift, **ES & EM** = Both Shifts.

Shift	Dataset	VLM-CL	VLM-RM	CLIP-FT	GVL-0S	GVL-1S	CLIP-GRU	VITA
ID	tk_pnp	0.038	0.029	0.251	0.269	0.252	<u>0.773</u>	0.782
ES	lm_pnp	0.017	0.033	0.149	0.305	0.272	<u>0.676</u>	0.725
	td_fold	0.031	0.072	0.152	0.326	0.318	<u>0.674</u>	0.709
	ft_fold	0.108	0.099	0.162	0.331	0.387	0.693	<u>0.658</u>
	rd_fold	0.095	0.055	0.126	0.372	0.406	0.726	<u>0.606</u>
	ms_sweep	-0.129	-0.226	0.148	0.158	0.150	<u>0.434</u>	0.490
EM	dt_tk_pnp	0.042	-0.041	0.149	0.258	0.211	0.856	<u>0.820</u>
	dt_tk_stack	0.035	0.046	0.099	0.254	0.277	<u>0.667</u>	0.708
ES & EM	dt_ft_stack	0.026	0.028	0.049	0.212	0.249	<u>0.674</u>	0.698
	dt_rd_pnp	0.023	0.041	0.211	0.329	0.316	0.747	<u>0.695</u>

4.3 EVALUATING GENERALIZATION UNDER DISTRIBUTION SHIFTS

We assess how well progress estimators generalize to novel environments, tasks, and robot embodiments, using subsets of BridgeData V2 curated to introduce broad variation along each of these axes. Environment shifts involve changes in scene layout or background: for example, `lm_pnp` is a pick-and-place task in front of a laundry machine, while `td_fold`, `ft_fold`, and `rd_fold` feature cloth folding on a variety of surfaces. `ms_sweep` adds a long-horizon sweeping task in a confined tray. Embodiment shifts are evaluated with the DeepThought robot, which differs from the training embodiment (WidowX 250) in both morphology and camera perspective. `dt_tk_pnp` (pick-and-place) and `dt_tk_stack` (stacking) retain the in-distribution environment under a new embodiment, while `dt_ft_stack` (stacking) and `dt_rd_pnp` (drawer pick-and-place) combine embodiment and environment shifts. Each dataset comprises 200 expert trajectories with task descriptions, except `ms_sweep`, which contains 100 owing to its limited size. Appendix B gives a full description of our evaluation datasets. We measure the performance of a value function estimator with the *Value Order Correlation* (VOC) metric (Ma et al., 2024), a general measure of the alignment between predicted progress toward task completion and the chronological order of frames in a visual trajectory. Formally, let $p_1, \dots, p_T$ denote the predicted progress values for a trajectory of T frames, and let $r_t = t$ represent the temporal index. VOC is defined as the Spearman rank correlation ρ_s between the predicted values and the temporal indices.

The results in Table 1 show that both VITA and CLIP-GRU consistently outperform GVL-0S and GVL-1S across the full suite of shifts, underscoring the limitations of pre-trained autoregressive VLMs and the broad importance of preserving temporal order for effective progress estimation. While both estimate progress effectively, VITA outperforms CLIP-GRU on the majority of evaluation datasets (6 out of 10). Both VITA and CLIP-GRU surpass the CLIP-based baselines that lack temporal modeling (CLIP-FT, VLM-CL, VLM-RM), reflecting the broad benefit of encoding temporal history for progress estimation. CLIP-FT matches GVL on the in-distribution task but fails to generalize to out-of-distribution settings. Both VLM-CL and VLM-RM perform poorly, likely because they lack temporal modeling. For GVL-0S and GVL-1S, in-context examples help in some cases and provide no benefit—or even degrade performance—in others. Both GVL methods do well on folding tasks (`td_fold`, `ft_fold`, `rd_fold`) but fall short on stacking (`dt_tk_stack`, `dt_ft_stack`) and pick-and-place (`dt_tk_pnp`, `lm_pnp`) tasks, suggesting that the autoregressive VLM used in GVL may be biased toward folding-like manipulations. In contrast, VITA performs consistently across every task type and distribution shift, pointing to broader generalization than in-context learning affords.

All methods except CLIP-FT degrade on the long-horizon task `ms_sweep`, yet VITA still attains the highest VOC score. In particular, VITA outperforms CLIP-GRU, indicating that encoding temporal history via test-time adaptation generalizes to long-horizon manipulation more effectively than modeling it through recurrent hidden states. Under embodiment shifts, VITA again generalizes strongly. On `dt_tk_pnp`, which uses a different robot embodiment but shares the environment and tasks of the training set, both VITA and CLIP-GRU exceed their in-distribution performance,

suggesting that learned temporal context transfers broadly across robot embodiments. In contrast, GVL-1S performs poorly on both `dt_tk_pnp` and `dt_tk_stack`, indicating that few-shot learning does not generalize under embodiment shifts in our evaluation setup.

4.4 DIFFERENTIATING EXPERT FROM NON-EXPERT TRAJECTORIES

Beyond generalization, we probe a second capability expected of a general value function: robustness in distinguishing expert from suboptimal visual trajectories, assigning lower progress scores to the latter. To this end, we compare model predictions on expert and scripted (non-expert) visual trajectories collected from the same in-distribution setting of BridgeData V2. The scripted trajectories are generated in the ToyKitchen environment with a heavily randomized controller (Walke et al., 2023), under the same embodiment and environmental configuration as the expert demonstrations. They span a variety of pick-and-place tasks whose object categories may overlap with the training set, including general objects (`sc_pnp_obj`), rigid items (`sc_pnp_robj`), soft toys (`sc_pnp_stoy`), utensils (`sc_pnp_uten`), and cloth (`sc_pnp_cloth`). Although these trajectories match the training distribution in task, embodiment, and environment, their behavior departs from the smooth, monotonic progress typical of expert demonstrations. Appendix C includes examples from the scripted dataset. Because scripted datasets are produced by a randomized controller, they are not designed to reflect quantifiable levels of suboptimality. Nonetheless, any effective value function estimator should assign lower VOC scores to suboptimal trajectories than to expert ones evaluated in the same setting. We measure discriminative success with *BinVOC*, a general binary metric defined as $1[\text{VOC}_{\text{exp}} > \text{VOC}_{\text{subopt}}]$, which is 1 when expert trajectories achieve higher VOC than suboptimal ones and 0 otherwise.

We summarize discriminative performance with *BINVOC*, averaged over all scripted datasets. Performance on expert visual trajectories (VOC_{exp}) is measured on the in-distribution test set `tk_pnp`, which features pick-and-place tasks akin to those in the scripted evaluation. VITA, GVL-0S, and GVL-1S all achieve perfect discrimination, consistently assigning higher VOC scores to expert trajectories across every task. VITA outperforms CLIP-GRU, suggesting that encoding temporal memory in recurrent hidden states is more prone to overfitting temporal shortcuts than the implicit memory formed by sequential test-time updates. The CLIP-based baselines (VLM-CL, VLM-RM) likely underperform because they lack temporal modeling altogether. CLIP-FT is more reliable but fails on one task, pointing to sensitivity to object distribution shifts.

4.5 ZERO-SHOT REWARD SHAPING FOR OFFLINE RL

We further evaluate our value function estimator on the Meta-World MT10 benchmark (Yu et al., 2020), a suite of ten diverse robotic manipulation tasks that includes pick-and-place, door opening, and pushing. This experiment tests whether VITA’s zero-shot reward shaping transfers across the sim-to-real boundary to offline RL: VITA is trained solely on real-world robotic data and applied zero-shot in this simulated setting. For each MT10 task, we generate 20 expert visual demonstrations with Meta-World’s expert policies, and the goal-conditioned value estimator then defines a dense reward for each visual observation. Policies are trained with Implicit Q-Learning (IQL) (Kostrikov et al., 2021) in the offline RL setting. We repeat expert data generation and policy training across 10 random seeds. After training, each multi-task policy is evaluated on 20 rollouts per task, and we compute the average return across episodes for each (task, seed) pair. Following the evaluation protocol of Agarwal et al. (2021), we report the interquartile mean (IQM) across all pairs together with 95% stratified bootstrap confidence intervals. The training details and evaluation protocol are described further in Appendix E.

Table 2 reports performance on the MT10 benchmark for the baselines under the evaluation protocol described above. VITA attains the highest IQM return (0.815), outperforming all CLIP-based baselines. A direct comparison with GVL is infeasible at this scale, since it relies on Gemini-1.5, a proprietary autoregressive VLM whose inference cost is prohibitive in RL settings, and open-source alternatives did not yield reliable progress estimates in our preliminary tests. We also report the fuzzy-logic dense reward provided by Meta-World (META-WL), which reaches 0.779. Despite its strong performance, META-WL is surpassed by VITA, indicating that a value estimator trained purely on real-world data can generalize effectively across the sim-to-real gap to simulated reward shaping.

Method	BinVOC
VLM-CL	0.40
VLM-RM	0.00
CLIP-FT	0.80
GVL-0S	1.00
GVL-1S	1.00
CLIP-GRU	0.80
VITA	1.00

(a) Expert vs. Non-Expert (BinVOC).

Method	IQM	95% CI
VLM-CL	0.760	[0.722, 0.791]
VLM-RM	0.746	[0.718, 0.771]
CLIP-FT	0.785	[0.759, 0.809]
CLIP-GRU	0.777	[0.734, 0.814]
META-WL	0.779	[0.750, 0.804]
VITA	0.815	[0.785, 0.838]

(b) Offline RL on MT10 (IQM).

Table 2: (a) Success in distinguishing expert from scripted robot demonstrations, measured by average BinVOC across the 5 in-distribution scripted datasets. (b) Offline RL performance on the Meta-World MT10 benchmark, measured by interquartile mean (IQM) with 95% stratified bootstrap CIs, illustrating transfer of a real-world-trained estimator to a simulated multi-task setting.

4.6 ABLATION STUDIES

4.6.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

We empirically validate the proposed dissimilarity-based sampling by comparing how different sub-trajectory sampling strategies affect VITA’s ability to distinguish expert from non-expert demonstrations. Full-trajectory sampling (Sun et al., 2024) computes the adaptation loss over all overlapping sub-trajectories with stride $s = 1$, which in our setup encourages overfitting to global temporal shortcuts. Random sampling increases diversity by selecting sub-trajectories uniformly, but does not target semantic diversity. Dissimilarity-based sampling instead explicitly promotes semantic diversity and, as a result, generalizes better. Our proposed method, which selects sub-trajectories that maximize pairwise dissimilarity, outperforms both full-trajectory and random sampling at differentiating expert from non-expert visual trajectories. Further details appear in Appendix F.1.

4.6.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

We analyze how different test-time adaptation strategies for incorporating temporal context affect value function estimation. VITA performs step-by-step adaptation without reset (implicit memory), incrementally updating parameters from θ_{t-1} . This lets the value function estimator encode trajectory history directly into its parameters, capturing both temporal order and accumulated temporal context. We compare VITA against three alternatives: (i) *trajectory-level adaptation (TTT-TR)*, which applies a single gradient update computed over the full trajectory; this batched update averages the loss across all frames and thereby discards temporal order; (ii) *memoryless adaptation (TTT-RS)*, which resets the adaptation module to θ_0 at every step and, by adapting only to the current observation, carries no temporal information forward; (iii) *explicit memory (TTT-EX)*, which also resets the adaptation module at every step but updates over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$). Across all datasets, VITA consistently outperforms the trajectory-level, memoryless, and explicit-memory variants, showing that incrementally accumulating temporal context is broadly the most effective strategy for value function estimation. The full analysis appears in Appendix F.2.

5 RELATED WORK

5.1 VLMS AS VALUE FUNCTION ESTIMATORS

Contrastive VLMs have been used broadly as zero-shot reward models (Baumli et al., 2023; Rocamonde et al., 2024) and goal-conditioned value functions (Ma et al., 2023b; 2024), relying on frame-level similarity scores but lacking temporal modeling. Some works fine-tune CLIP on domain-specific video datasets (Fan et al., 2022; Jiang et al., 2024) to produce dense rewards, requiring supervision that narrows their zero-shot applicability. RoboCLIP (Sontakke et al., 2023) enables one-shot reward specification without fine-tuning, but yields sparse rewards and still requires demonstrations. Large-scale multimodal pre-training approaches (Alakuijala et al., 2024; Li et al., 2024; Ma et al., 2023b) learn language-conditioned value functions by leaning on scale to achieve generalization.

ReWiND (Zhang et al., 2025b) trains a cross-modal sequential transformer on augmented large-scale data with video rewinding, combined with a few in-domain demonstrations to broaden generalization. In contrast, VITA adapts a pre-trained contrastive vision–language encoder at inference, capturing both temporal and semantic context through test-time adaptation, without any large-scale multimodal pre-training or additional domain-specific demonstrations. Because it is agnostic to the underlying encoder, VITA applies broadly: it can be layered on top of any pre-trained vision–language representation that encodes a universal value function, enhancing its generalization and temporal reasoning through meta-learned test-time adaptation. Autoregressive VLMs (Alayrac et al., 2022) exploit in-context learning over trajectories and have been applied across settings such as success detection (Du et al., 2023) and goal-conditioned value functions (Ma et al., 2024). GVL (Ma et al., 2024) mitigates the monotonic bias of ordered pre-training data by shuffling frames at inference, but this discards temporal order—information that is essential for temporal reasoning and for distinguishing visually similar yet temporally distinct states. Our approach instead preserves chronological order while capturing temporal context through test-time adaptation, avoiding such shortcuts.

5.2 TEST-TIME ADAPTATION

Test-time adaptation methods offer parameter-efficient ways to adapt VLMs across tasks without full fine-tuning. Prompt-based methods optimize model parameters at test time from the task context, although their underlying mechanisms remain opaque (Zhou et al., 2022; Khattak et al., 2023; Ma et al., 2023a). Lim et al. (2025) argued that CLIP representations can broadly support test-time adaptation, motivated by the observation that CLIP encodes human-interpretable concepts discovered via mechanistic interpretability (Bricken et al., 2023). CLIP-based similarity has also served as a reward signal, with parameters optimized at test time via reinforcement learning (Zhao et al., 2024). A further, broadly applicable approach is test-time training (Sun et al., 2020), in which an adaptation module is updated at each timestep via a self-supervised loss. Unlike meta-reinforcement learning methods (Finn et al., 2017; Bauer et al., 2023), which require task-specific adaptation episodes during training, test-time training enables direct online adaptation at inference across arbitrary tasks, without task labels.

6 DISCUSSION, LIMITATIONS, AND FUTURE WORK

6.1 DISCUSSION

We show that test-time adaptation enables value functions for robotic manipulation to generalize broadly across distribution shifts in task, environment, and embodiment. Beyond this, VITA distinguishes expert from non-expert visual trajectories and performs zero-shot reward shaping, opening the door to a range of downstream applications in RL and imitation learning. By updating sequentially over a trajectory, VITA encodes history into its parameters, capturing temporal and semantic context more effectively than CLIP-based baselines and than in-context learning with autoregressive VLMs. VITA outperforms CLIP-GRU on both expert–non-expert discrimination and offline RL, indicating that implicit memory formed via sequential test-time updates generalizes more effectively than explicitly encoding temporal history in recurrent hidden states for zero-shot value function estimation. Pre-trained autoregressive VLMs (GVL), for all their general capabilities, are not explicitly trained for progress estimation, which demands temporal reasoning over visual trajectories. In our setup, the inference-time overhead of per-timestep adaptation is negligible—only a lightweight adaptation module is updated at test time—so real-time applicability is preserved across the settings we study.

6.2 LIMITATIONS AND FUTURE WORK

Across our experiments, test-time adaptation improves the generalization of zero-shot progress estimation to a broad range of unseen tasks and environments, though challenges can remain in settings with high execution variability or extended durations. While the adaptation overhead is negligible thanks to the lightweight adaptation module, updating a value function estimator at every timestep could be unsafe in some deployments, which may limit applicability in the most demanding real-time scenarios. Looking ahead, we plan to explore alternative test-time adaptation schemes for vision–language models, particularly for training agents within world models. Extending VITA to real-time closed-loop control and more complex RL environments is likewise a promising direction.

Because it is agnostic to the underlying vision–language encoder, VITA can in principle enhance the generalization and temporal reasoning of any domain-specific pre-trained multimodal representation through meta-learned test-time adaptation, suggesting applicability well beyond the domains studied here. Finally, although we provide empirical evidence that dissimilarity-based sampling mitigates shortcut learning, a theoretical account of how diversity-based sampling shapes shortcut learning remains open. This lies beyond the scope of our core contribution—a value function estimator that meta-learns a general test-time adaptation mechanism to improve zero-shot performance and temporal reasoning.

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A TRAINING DATASET

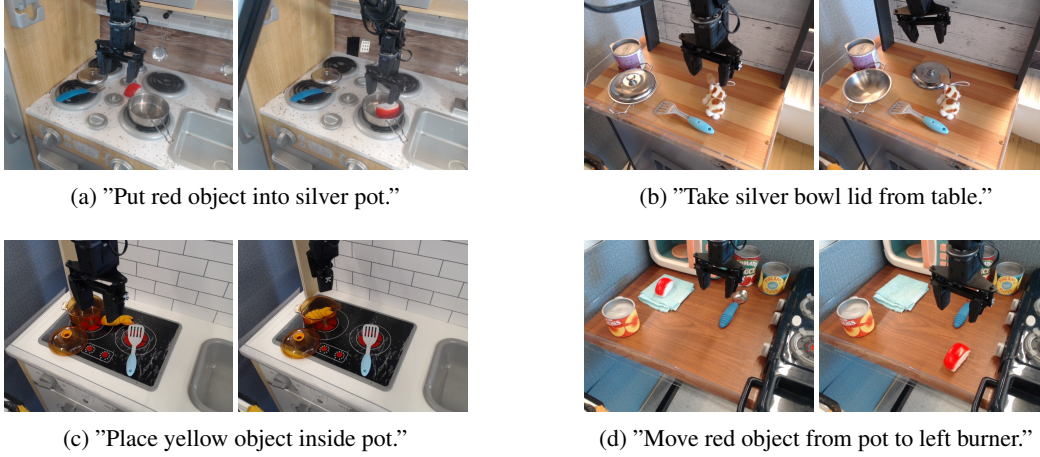


Figure 4: Each subfigure shows the start and end frames of an expert demonstration used for training, together with its natural language task description. The demonstrations span four distinct ToyKitchen environments.

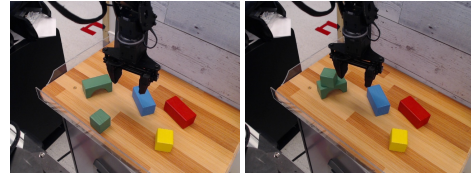
B EVALUATION DATASETS

Table 3: Descriptions of the evaluation datasets, spanning a range of task types, environments, and embodiments.

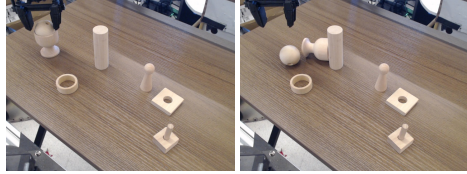
Dataset	Task Type	Environment	Embodiment
tk_pnp	pick-and-place	toy kitchen	WidowX 250
lm_pnp	pick-and-place	laundry machine	WidowX 250
td_fold	fold cloth	tabletop (dark wood)	WidowX 250
ft_fold	fold cloth	folding table	WidowX 250
rd_fold	fold cloth	robot desk	WidowX 250
ms_sweep	sweep	folding table (tray)	WidowX 250
dt_tk_pnp	pick-and-place	toy kitchen	DeepThought
dt_tk_stack	stack blocks	toy kitchen	DeepThought
dt_ft_stack	stack blocks	folding table	DeepThought
dt_rd_pnp	pick-and-place	robot desk (drawer)	DeepThought



(a) dt_tk_ms: "Move yellow object from burner to center."



(b) dt_tk_stack: "Place green cube on top of arch."

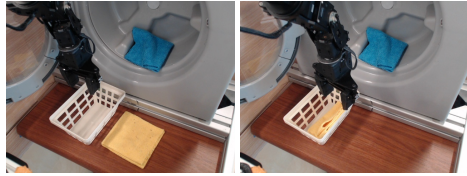


(c) dt_ft_stack: "Move egg to table."



(d) dt_rd_pnp: "Put orange object in drawer."

Figure 5: Each subfigure shows the start and end frames of an evaluation trajectory under embodiment shift, together with its natural language task description. The top row depicts tasks in the same environment (ToyKitchen) with a different robot (DeepThought), while the bottom row spans tasks that additionally involve new environments.



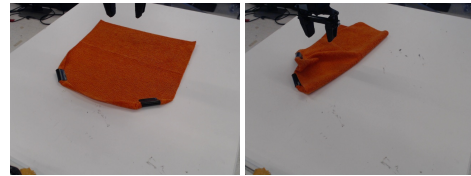
(a) lm_pnp: "Place blue cloth inside washer."



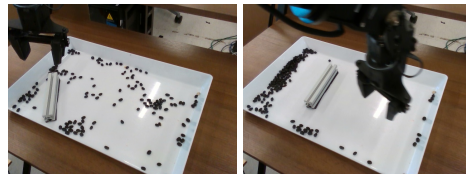
(b) td_fold: "Fold cloth bottom right to top left."



(c) ft_fcnp: "Fold clothes from left to right."



(d) rd_fold: "Fold cloth bottom right to top left."



(e) ms_ft_sweep: "Sweep into pile."

Figure 6: Each subfigure shows the start and end frames of an evaluation trajectory under environment shift, together with its natural language task description. All tasks use the same robot embodiment but span visually and structurally diverse environments.

C SCRIPTED DATASET



(a) `sc_pnp_obj`: "Pick up an object and place it."



(b) `sc_pnp_robj`: "Pick up a rigid object and place it."



(c) `sc_pnp_stoy`: "Pick up a soft object and place it."



(d) `sc_pnp_uten`: "Pick up a kitchen utensil and place it."



(e) `sc_pnp_cloth`: "Move the cloth."

Figure 7: Each subfigure shows the start and end frames of a scripted evaluation trajectory collected in the ToyKitchen environment with the WidowX 250 robot. All tasks involve pick-and-place behavior across a range of object categories. We use generic task descriptions in the absence of publicly available annotations.

D MODEL ARCHITECTURE

D.1 MODEL AND TRAINING DETAILS

We adopt the OpenCLIP ViT-B/32 encoder pre-trained on the OpenAI dataset as a frozen backbone. Each visual observation and task description is encoded with the CLIP vision and text encoders, respectively. We chose a joint (concatenation-based) representation over an element-wise product based on improved validation performance. The adaptation module f_{adapt} is a two-layer residual MLP with GELU activation and projection dimension $d' = 64$. The model is trained with the AdamW optimizer using a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , and a cosine learning rate schedule with 10% warmup. We use a batch size of 32 and pad all trajectories to a maximum length of 120 frames (matching the longest sequence in the training set). The weighting coefficient λ_{self} of the self-supervised loss in the total training objective is set to 0.5, selected on validation performance. We train for 5 epochs, since validation VOC typically plateaus early and further training yields no improvement. For dissimilarity-based sampling, we set the stride to $s = 1$, the sub-trajectory length to $w_{\text{tr}} = 8$, and the number of selected windows to $k = 8$, unless otherwise specified. All experiments ran on NVIDIA RTX 6000 Ada Generation GPUs.

D.2 TEST-TIME TRAINING HYPERPARAMETERS

At inference time, we adapt only the temporal adaptation module f_{adapt} , keeping the same projection dimension ($d' = 64$) as in training. We perform a single gradient step ($t_{\text{ep}} = 1$) with a learning rate $\eta = 0.1$. This configuration was chosen from a sweep over learning rates

Table 4: Effect of the test-time learning rate η and the number of adaptation steps t_{ep} on VITA’s validation performance, measured by VOC.

η	$t_{\text{ep}} = 1$	$t_{\text{ep}} = 5$	$t_{\text{ep}} = 10$
0.001	0.225	0.345	0.453
0.01	0.454	0.744	0.781
0.05	0.746	0.670	0.515
0.1	0.783	0.512	0.369
1.0	0.299	0.183	0.181
5.0	0.169	0.182	0.157

$\{5.0, 1.0, 0.1, 0.05, 0.01, 0.001\}$ and adaptation steps $\{1, 5, 10\}$, based on the validation performance reported in Table 4. The results show that both (i) a single update with a larger learning rate ($\eta = 0.1$, $t_{\text{ep}} = 1$) and (ii) multiple updates with a smaller learning rate ($\eta = 0.01$, $t_{\text{ep}} = 10$) achieve comparable validation performance under test-time training. We adopt the former, as it requires fewer adaptation steps while performing slightly better.

D.3 BASELINES

VITA uses $t_{\text{ep}} = 1$, projection dimension $d' = 64$, and learning rate $\eta = 0.1$, selected via hyperparameter tuning as described above. CLIP-FT follows VITA’s architecture but drops the adaptation module and self-supervised loss: it applies a frozen CLIP encoder followed by a linear projection and a two-layer MLP to predict task progress. It is trained with standard supervised regression, without meta-learning or test-time adaptation. To give it greater expressivity, it uses an $8\times$ larger projection matrix and $10\times$ more training steps than our method (Lin et al., 2022). For VLM-RM, we use a baseline prompt that describes the environment. CLIP-GRU likewise follows VITA’s architecture (frozen CLIP encoder and regression head h) but replaces the test-time adaptation module f_{adapt} with a Gated Recurrent Unit (GRU) that encodes temporal history explicitly through recurrent hidden states; the GRU is single-layer with hidden size 64, and its final hidden state is passed to the regression head h . For both GVL-0S and GVL-1S, we use the latest release of Gemini 1.5 Pro, `gemini-1.5-pro-latest` (Gemini Team et al., 2024), whereas the original GVL implementation used the earlier `gemini-1.5-pro`. To gauge how broadly the GVL recipe holds across models, we also evaluated it with the open-source autoregressive VLM Qwen-VL 2.5 (Yang et al., 2024), which failed to overcome temporal bias despite frame shuffling, while GPT-4o (Hurst et al., 2024) consistently declined to produce scalar progress estimates in our setup. All evaluations followed the prompt template introduced in the original GVL work (Ma et al., 2024).

E OFFLINE RL SETUP

E.1 TRAINING CONFIGURATION

We use Implicit Q-Learning (IQL) with a single configuration kept fixed across all tasks. Expectile regression is set to $\tau = 0.7$, with advantage weighting temperature 3.0 and clipping threshold 100.0. The actor and critic use learning rates 1×10^{-4} and 3×10^{-4} , respectively, with batch size 256. Policies are trained for 100,000 gradient steps, with evaluation every 10,000 steps on 20 rollouts per task (horizon 150). Implementation follows `d3rlpy`, with configuration: `expectile=0.7`, `weight_temp=3.0`, `max_weight=100.0`, `actor_lr=1e-4`, `critic_lr=3e-4`, `batch_size=256`.

E.2 EVALUATION PROTOCOL

Following Agarwal et al. (Agarwal et al., 2021), we report the interquartile mean (IQM) of returns across all task–seed pairs. The IQM averages the middle 50% of scores, reducing sensitivity to outliers relative to the mean or median. For statistical robustness, we estimate 95% confidence intervals with a stratified bootstrap of 10,000 resamples, stratified by task. This protocol is applied uniformly to every method in the Meta-World MT10 experiments.

F ABLATION STUDIES

F.1 EFFECT OF DISSIMILARITY-BASED SAMPLING ON DISCRIMINATIVE PERFORMANCE

Table 5: Disc@VOC across the scripted datasets under a range of training-time sampling strategies for test-time adaptation. A checkmark marks successful discrimination between expert and suboptimal trajectories. The sampling methods **FT** (full-trajectory), **Rand** (random sampling), and **Diss** (dissimilarity-based sampling) are parameterized by (w_{tr}, k) .

Dataset	FT (8,-)	Rand (8,8)	Diss (8,4)	Diss (4,8)	Diss (8,8)
sc_pnp_obj		✓	✓		✓
sc_pnp_robj	✓	✓	✓	✓	✓
sc_pnp_stoy			✓		✓
sc_pnp_uten	✓	✓	✓	✓	✓
sc_pnp_cloth		✓	✓	✓	✓
Avg. Disc@VOC	0.20	0.8	1.0	0.60	1.0

We empirically validate the proposed dissimilarity-based sampling by comparing how different training-time sub-trajectory sampling strategies affect our model’s ability to discriminate expert from suboptimal trajectories. In Table 5, we compute Disc@VOC over the 5 scripted datasets, capturing whether a model assigns a higher average VOC to expert trajectories (tk_pnp) than to each evaluation dataset. We contrast three strategies: (1) full-trajectory sampling adapts over all overlapping sub-trajectories of length w_{tr} with stride s , following prior work (Sun et al., 2024); (2) random sampling draws k sub-trajectories of length w_{tr} uniformly per video; and (3) our proposed dissimilarity-based sampling builds a candidate set of length- w_{tr} sub-trajectories with stride s and selects the k that maximize pairwise dissimilarity in the representation space.

The results show that dissimilarity-based sampling with $w_{tr} = 8$ and $k = 8$ yields perfect discriminative performance. Sampling fewer windows ($k = 4$) at size $w_{tr} = 8$ matches this, while shortening the sub-trajectory length to $w_{tr} = 4$ reduces effectiveness. The full-trajectory baseline performs worst, confirming that adapting over all consecutive windows tends to overfit to global temporal cues. Random sampling improves on full-trajectory sampling but remains less reliable than dissimilarity-based selection. Taken together, these findings underscore that both sub-trajectory diversity and semantic locality are broadly important for discriminative test-time adaptation.

F.2 EFFECT OF TEMPORAL MEMORY ON VALUE FUNCTION ESTIMATION

Table 6: Validation VOC scores for progress estimation across the full range of distribution shifts. ID = In-Distribution, ES = Environment Shift, EM = Embodiment Shift, ES+EM = Environment and Embodiment Shift.

Shift	Dataset	TTT-TR	TTT-RS	TTT-EX	VITA
ID	tk_pnp	0.1917	0.1918	0.1957	0.7822
ES	lm_pnp	0.1843	0.1854	0.1826	0.7246
	td_fold	0.1402	0.1393	0.1322	0.7085
	ft_fold	0.1302	0.1311	0.1683	0.6583
	rd_fold	0.1161	0.1172	0.1154	0.6056
	ms_sweep	-0.0225	-0.0191	0.0102	0.4898
EM	dt_tk_pnp	0.2123	0.2118	0.2069	0.8203
	dt_tk_stack	0.0833	0.0830	0.0747	0.7081
ES+EM	dt_ft_stack	0.0558	0.0537	0.0439	0.6979
	dt_rd_pnp	0.1665	0.1660	0.1559	0.6951

We evaluate four adaptation strategies: (1) TTT-TR, which performs a single gradient update computed over the full trajectory (offline); this batched update averages the loss across all frames, discarding temporal order; (2) TTT-RS, which resets the adaptation module at every step and updates using only the current visual observation (memoryless), carrying no temporal information forward; (3) TTT-EX, which resets the adaptation module at every step and updates over a local window of recent frames matching the meta-training window size ($w_{tr} = 8$), averaging gradients over the window and thus discarding the temporal ordering within it; (4) VITA (OURS), which updates the adaptation module incrementally at every timestep (implicit memory), forming an implicit memory through the sequential accumulation of temporal history in the adapter parameters. We ran the same test-time hyperparameter sweeps for every variant, as described in Appendix D. We found that VITA, TTT-RS, and TTT-TR perform best with a learning rate of 0.1 and a single adaptation step, whereas TTT-EX performs best with a learning rate of 1.0 and one adaptation step.

The results in Table 6 show that TTT-TR, TTT-RS, and TTT-EX perform comparably. This indicates that batched updates—whether over a short explicit window (TTT-EX) or the full trajectory (TTT-TR)—behave much like the memoryless baseline (TTT-RS). In contrast, VITA consistently outperforms all three variants across the full range of shifts, reinforcing the broad importance of sequential, per-timestep updates for effective value function estimation in real-world robotic manipulation. Under implicit memory (VITA), updating over a local window of frames would incorporate the most recent observations twice—explicitly through the window and implicitly through the accumulated parameters—while updating less frequently (non-sequentially) would lose information by skipping intermediate frames. The implicit-memory test-time training paradigm therefore calls for sequential, per-timestep updates without reset.

G COMPLEXITY ANALYSIS OF DISSIMILARITY-BASED SAMPLING

Let $\mathcal{W}$ denote the candidate set of size $N_c = |\mathcal{W}|$. To approximate diverse subset selection without the exponential cost of enumerating all $\binom{N_c}{b}$ subsets as described in Eq. 4, we adopt a scoring-based heuristic that assigns each candidate window a diversity score equal to the sum of its pairwise distances to all others (i.e., the row sum of the distance matrix). With the number of candidate windows given by $N_c = T - w_{tr} + 1$, each represented by a flattened feature vector of dimension $m = w_{tr} \cdot d$ where d is the dimension of each multimodal representation z_t , we compute the full pairwise distance matrix $D \in \mathbb{R}^{N_c \times N_c}$ on the GPU with batched `cdist`, at a cost of $\mathcal{O}(N_c^2 \cdot w_{tr} \cdot d)$ operations.

In our experiments, training trajectories are short (mean length 69, maximum length 120). With a batch size of 32, feature dimension $d = 1024$ (the CLIP-based multimodal representation), and window length $w_{tr} = 8$, the worst-case computational cost is:

$$32 \cdot N_c^2 \cdot w_{tr} \cdot d = 32 \cdot 114^2 \cdot 8 \cdot 1024 \approx 340 \text{ MFLOPs.}$$

This overhead is negligible relative to the cost of a single forward and backward pass through our model, yet it reliably promotes the selection of diverse windows across trajectories of varying length.